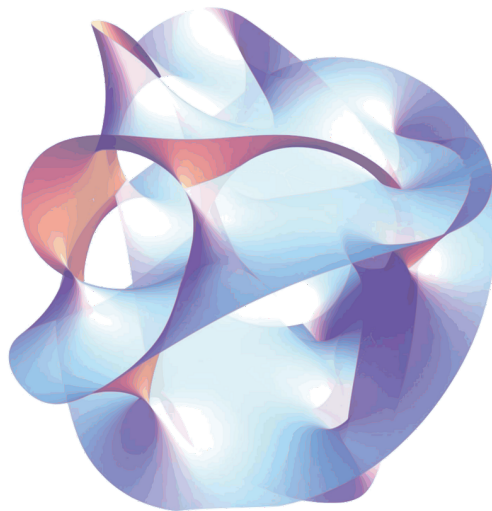


Dimensional reduction in supergravity and duality symmetries

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Notations and abbreviations

We will during this internship use the vielbein formalism. We will use the same notation as in [1] which is

- Greek hatted letter for curved indices ($\hat{\mu}, \hat{\nu}, \hat{\rho}, \dots$)
- Latin hatted letter for flat indices ($\hat{p}, \hat{m}, \hat{n}, \dots$)
- For space-time indices taking D values, late greek letters are used in the curved case ($\mu, \nu, \rho, \dots$) and Latin letters in the flat case (p, m, n)
- For internal indices taking E values, early Greek letters in the curved case ($\alpha, \beta, \gamma, \dots$) and Latin letters in the flat case (a,b,c)

The general coordinate $x^{\hat{\mu}}$ will thereofre be made of space time coordinate x^{μ} and internal coordinate y^{α} . The flat lorenz

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1 Introduction

Since the 19th-century unification of electricity and magnetism, and the later development of classical and quantum field theory, physicists have been looking for a single mathematical framework that could describe all fundamental interactions at once. A lot of progress has been made for the electromagnetic, weak and strong forces, which are now successfully unified in the Standard Model. Gravity, however, remains the main stumbling block. Unlike the other forces, it resists quantization in a straightforward way, and its description by General Relativity [2] sits completely apart from the quantum field-theoretic language we use for everything else. One of the earliest attempts to change this was proposed by Kaluza in 1921 [3] and later refined by Klein:

by adding one extra spatial dimension compactified on a small circle, five-dimensional General Relativity naturally reduces to four-dimensional Einstein gravity coupled to electromagnetism. Even if the original theory had experimental failure, this Kaluza-Klein mechanism opened a whole new way of thinking about unification, and it is now at the heart of modern approaches like supergravity [4] and string theory.

Another important part of physics since the beginning has been the discovery of symmetry, indeed thanks to symmetry we can greatly reduce the level of difficulty of some theory, and in other fields we can even show that some multiple theories describe in fact the same universe, but with a slightly different description, this has played a big part in string theory, and supergravity, or any advanced theory.

In this internship we will begin by describing what is a field theory to seek how to describe the different action of the field we will be using, after that we will follow the steps of Kaluza and Klein by doing a dimensional reduction from D+E dimension to 4 dimension. However we will not just use the classical Einstein action, but the NS-NS action of supergravity which basically adds a new field, we will do this by using the vielbein formalism as it is useful for the description of fermions even here we will solely try to describe bosonic fields, then we will see that even if in high dimension only the $SL(E,E)$ dimension is detectable without too much work, that when we go into low dimension another $O(E,E)$ symmetry will appear, and that means that a lot of other theories are in fact equivalent to the one we are using.

2 Lagrangian field theory

2.1 Lagrangian density

In field theory, just as Newton's second law is "volumized" into the Navier-Stokes equations by distributing quantities over space, we replace discrete degrees of freedom with continuous field densities defined at every point in spacetime. Therefore we will rewrite the Lagrangian as the integral over space of a new quantity called Lagrangian density, which will often be referred to as just the Lagrangian. Thus we obtain this equation:

$$L = \iiint \mathcal{L}(\eta_\rho, \frac{\partial \eta_\rho}{\partial x^\nu}) dx dy dz \quad (1)$$

Now using Einstein's convention we can rewrite the action in a new light :

$$S = \int \mathcal{L} dx^\mu \quad (2)$$

2.2 General example for scalar and vectorial fields, and result for two forms

In field theory, the Lagrangian has a classical prescription for scalar and vector fields, but no general rule exists for other cases each field type requires its own treatment. For this work, only the scalar, vector, and two-form Lagrangians are needed, along with the metric, which is handled by Einstein's equations.

So for a scalar field we have :

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi \quad (3)$$

and for a vectorial field we have

$$\mathcal{L} = -\frac{1}{2} F_{\mu\nu} F^{\mu\nu} \quad (4)$$

where for the field A

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

In a vacuum the Einstein equations give us

$$\mathcal{L} = R$$

where R is Ricci scalar finally for the Lagrangian of a two-form field we will look in [5] where it is given by the equation :

$$\mathcal{L} = \frac{1}{12} g^{\hat{\mu}\hat{\mu}'} g^{\nu\nu'} g^{\rho\rho'} H_{\mu\nu\rho} H_{\mu'\nu'\rho'} \quad (5)$$

where for the two-form $B_{\nu\rho}$ we have

$$H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + \text{cyclic perm}$$

3 NS-NS Action Of supergravity

In this section we perform a Kaluza-Klein reduction on the NS-NS action which is

$$S = \int dx^D \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} e^{\hat{\phi}} (\hat{R} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi} - \frac{1}{12} g^{\hat{\mu}\hat{\mu}'} g^{\hat{\nu}\hat{\nu}'} g^{\hat{\rho}\hat{\rho}'} H_{\hat{\mu}\hat{\nu}\hat{\rho}} H_{\hat{\mu}'\hat{\nu}'\hat{\rho}'}) \quad (6)$$

where $\rho(E)$ is a normalisation factor, corresponding in practice to the invariant volume of the internal space.

And where $\hat{R}$ is similar as the classical Hilbert-Einstein action but in $D+E$ dimension, $\hat{\phi}$ is a dilaton field and $-\frac{1}{12} H_{\hat{\mu}\hat{\nu}\hat{\rho}} H^{\hat{\mu}\hat{\nu}\hat{\rho}}$ is the action of a two form. Finally as no field depend on internal component and knowing what $\rho(E)$ we can just simplify this integral. Now we will separate this action into 3 part one for each action so we have We consider an universe in $D+E$ dimensions, with $D=4$, our goal will be to determine how an observer confined to the D -dimensional world perceives the extra dimensions.

3.1 Hypothesis

We assume that none of the fields depend on the internal coordinates, i.e. for any field $\partial_{\alpha} A^{\hat{\mu}} = 0$. This mean that the reduction will be done on a E-d torus.

3.2 different part of the action

We can separate the action into two main component one being the gravitational part with the dilaton and the otherone being the two forms part so we have:

$$S_{\hat{g}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} [\hat{R} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi}], \quad (7a)$$

$$S_{\hat{B}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} - \frac{1}{12} H_{\hat{\mu}\hat{\nu}\hat{\rho}} H^{\hat{\mu}\hat{\nu}\hat{\rho}}, \quad (7b)$$

4 gravitational action reduction

We will begin with the gravitational part of the NS-NS action.

4.1 Rewriting the action

For this part we don't need the $\hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi}$ part of the action so we will fully ignore it. Now using the fact that $\hat{R}$ can be written as

$$\hat{R} = \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}}$$

where

$$\hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} = \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{\nu}\hat{t}\hat{s}} - (\mu \leftrightarrow \nu)$$

and the ω are defined By

$$\hat{\omega}_{\hat{t}\hat{r}\hat{s}} = -\hat{\Omega}_{\hat{t}\hat{r}\hat{s}} + \hat{\Omega}_{\hat{r}\hat{s}\hat{t}} - \hat{\Omega}_{\hat{s}\hat{t}\hat{r}}$$

and finally the Ω are defined thanks to the vielbein with

$$\hat{\Omega}_{\hat{r}\hat{s}\hat{t}} = \frac{1}{2} (\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} - \hat{V}_{\hat{r}}^{\hat{\nu}} \hat{V}_{\hat{s}}^{\hat{\mu}}) \partial_{\hat{\nu}} \hat{V}_{\hat{\mu}\hat{r}}$$

with all of that we are going to rewrite the action $S_{\hat{g}}$ in terms of $\hat{\omega}$

$$\begin{aligned} S_{\hat{g}} &= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{R} \\ &= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} \\ &= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} ([\partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} - \partial_{\hat{\nu}} \hat{\omega}_{\hat{\mu}\hat{r}\hat{s}}] + [\hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{\nu}\hat{t}\hat{s}} - \hat{\omega}_{\hat{\nu}\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{\mu}\hat{t}\hat{s}}]) \\ &= \int dx^D \{ 2\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{s}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{s}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}}] \} \\ &= \int dx^D \{ \hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{s}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{s}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} \partial_{\hat{\mu}} (\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}}) \} \\ &= \int dx^D \{ \hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{s}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{s}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} [\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \partial_{\hat{\mu}} \hat{V}^{\hat{\nu}\hat{s}} + \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{V}^{\hat{\mu}\hat{r}} + e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{V} + \hat{V} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} e^{-\hat{\phi}}] \} \\ &= \int dx^D \hat{V} e^{-\hat{\phi}} \{ \hat{\omega}_{\hat{s}}^{\hat{t}} \hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} + \hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{s}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{r}} + 2\hat{\omega}_{\hat{s}}^{\hat{r}} \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi} \} \end{aligned} \quad (8)$$

4.2 Vielbein Anstaz

Now that we have explicitly written the action in terms of either latin connexion and or the two other fields, what we have to do is to suppose an ansatz for the vielbein. Why the vielbein and not the metric you may ask? Because as stated in ??, all the information of the metric can be contained within the vielbein, and as we primarily work with them for the action, it is easier to do an ansatz on them. Now we will do the same ansatzes as in the [1], however we will consider that γ they use is null since the beginning, and we will use the same parameter number as in [5]. Finally thanks to the local Lorentz invariance we can choose a triangular parametrisation giving us

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} V_{\mu}^r & A_{\mu}^{(1)\beta} \Phi_{\beta}^a \\ 0 & \Phi_{\alpha}^a \end{pmatrix} \quad (9)$$

Now still using (??) we can compute the inverse vielbein and then the metric and then the inverse metric that gave us

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} V_r^{\mu} & -A_r^{\alpha} \\ 0 & \Phi_a^{\alpha} \end{pmatrix}, \quad (10)$$

for the inverse vielbein, and

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g_{\mu\nu} - A_{\mu}^{\gamma} A_{\nu\lambda} & -A_{\mu\beta} \\ A_{\nu\alpha} & -h_{\alpha\beta} \end{pmatrix}, \text{ and } \hat{g}^{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g^{\mu\nu} & -A^{\mu\beta} \\ -A^{\nu\alpha} & -h^{\alpha\beta} + A_{\rho}^{\beta} A^{\rho\alpha} \end{pmatrix}, \quad (11)$$

for the metric

4.3 Gauge observation

Now that we have done the ansatz, we would like to know what each object are, for that we can use the classical gauge transformation $x'^{\mu} = x^{\mu} + \xi^{\mu}(x)$, and use the fact that it can be easily rewritten as a Lie derivative. Indeed we already know of different objects transform under Lie derivative so we would easily be able to identify what each object are.

So firstly let's list every known transformation:

$$\delta\phi(x) = \xi^{\mu} \partial_{\mu} \phi(x), \quad (12)$$

$$\delta U^{\mu} = \xi^{\rho} \partial_{\rho} U^{\mu} - (\partial_{\rho} \xi^{\mu}) U^{\rho}, \quad (13)$$

$$\delta\omega_{\mu} = \xi^{\rho} \partial_{\rho} \omega_{\mu} + (\partial_{\rho} \xi^{\mu}) \omega_{\rho}, \quad (14)$$

$$\delta T_{\nu}^{\mu} = \xi^{\rho} \partial_{\rho} T_{\nu}^{\mu} - (\partial_{\rho} \xi^{\mu}) T_{\nu}^{\rho} + (\partial_{\nu} \xi^{\rho}) T_{\rho}^{\mu}, \quad (15)$$

we now apply the form transformation to each block of the vielbein, in order to see whether it corresponds to some familiar type of field still thanks to (12). Let us start with the block $\hat{V}_{\mu}^r$:

$$\begin{aligned} \delta \hat{V}_{\mu}^r &= \delta V_{\mu}^r = \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}_{\mu}^r + \partial_{\mu} \xi^{\hat{\rho}} \hat{V}_{\hat{\rho}}^r \\ &= \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}_{\mu}^r + \partial_{\mu} \xi^{\hat{\nu}} \hat{V}_{\hat{\nu}}^r \\ &= \xi^{\rho} \partial_{\rho} V_{\mu}^r + \partial_{\mu} \xi^{\nu} V_{\nu}^r \end{aligned}$$

which is exactly the transformation law of a vector field under a Lie derivative [4]. Applying the same method to the other blocks, we obtain, for Φ_{α}^a :

$$\delta \Phi_{\alpha}^a = \xi^{\rho} \partial_{\rho} \Phi_{\alpha}^a$$

which is exactly the transformation law of a scalar field. Finally, for A_{μ}^{α} we obtain:

$$A_{\mu}^{\alpha} = \xi^{\rho} \partial_{\rho} A_{\mu}^{\alpha} + \partial_{\mu} \xi^{\rho} A_{\rho}^{\alpha} + \partial_{\mu} \xi^{\alpha}$$

which corresponds to the transformation law of a vector field, up to an extra correction term. One may further note that, restricting to ξ^{α} only, we obtain

$$\delta V_{\mu}^r = \delta \Phi_{\alpha}^a = 0, \quad (16a)$$

$$\delta A_{\mu}^{\alpha} = \partial_{\mu} \xi^{\alpha}. \quad (16b)$$

so that what remains of the D -dimensional diffeomorphisms, once restricted to ξ^{α} independent of the internal coordinates y , corresponds to a set of E abelian gauge symmetries. The vielbein in (9) thus contains the D -dimensional vielbein (i.e. the graviton), E vector fields A_{μ}^{α} , and E^2 scalar fields Φ_{α}^a . However, because of the local Lorentz invariance in the internal directions, only $\frac{1}{2}E(E+1)$ of these scalars actually correspond to propagating degrees of freedom. We also introduce $\delta = \det \Phi$

4.4 Ricci scalar reduction

Now that we better know the tool we will use, We can begin the calculation. For a starter we will do the Einstein action in D+E dimension twiked by the addition of the dilation.

Firstly let's begin by expending completly the $S_{\hat{g}}$, this will show us that we technically have to compute 8 ω , however thanks to the antisymetrie og the last to index we can reduce that to 6. So the action is

$$S = -\frac{1}{4\kappa^2} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \left[\hat{\omega}^a_{bc} \hat{\omega}^{bc}_a - \hat{\omega}^a_{tb} \hat{\omega}^{bt}_a + \hat{\omega}^a_{sc} \hat{\omega}^{sc}_a + \hat{\omega}^a_{st} \hat{\omega}^{st}_a - \hat{\omega}^r_{bc} \hat{\omega}^b_{r^c} - \hat{\omega}^r_{tb} \hat{\omega}^{bt}_r - \hat{\omega}^r_{sc} \hat{\omega}^s_{r^c} \right. \\ \left. \hat{\omega}^r_{st} \hat{\omega}^{st}_r + \hat{\omega}^a_{ac} \hat{\omega}^b_{b^c} + \hat{\omega}^a_{ta} \hat{\omega}^{bt}_b + \hat{\omega}^a_{ac} \hat{\omega}^s_{s^c} - \hat{\omega}^a_{ta} \hat{\omega}^s_{s^t} + \hat{\omega}^r_{rc} \hat{\omega}^b_{b^c} - \hat{\omega}^r_{rt} \hat{\omega}^{bt}_b + \hat{\omega}^r_{rc} \hat{\omega}^s_{s^c} + \hat{\omega}^r_{rt} \hat{\omega}^s_{s^t} + \right. \\ \left. 2\hat{\omega}_s{}^{rs} \hat{V}_r{}^\mu \partial_\mu \hat{\phi} + 2\hat{\omega}_b{}^{rb} \hat{V}_r{}^\mu \partial_\mu \hat{\phi} \right] \quad (17)$$

It then becomes clear that we only need to compute six components of ω , namely

$$\hat{\omega}_{pmn}, \hat{\omega}_{pma}, \hat{\omega}_{pab}, \\ \hat{\omega}_{cmn}, \hat{\omega}_{cma}, \hat{\omega}_{cab},$$

One can also check that computing these six components only requires the following eight components of Ω :

$$\hat{\Omega}_{pmn}, \\ \hat{\Omega}_{pma}, \hat{\Omega}_{pam}, \hat{\Omega}_{apm}, \\ \hat{\Omega}_{mab}, \hat{\Omega}_{amb}, \hat{\Omega}_{abm}, \\ \hat{\Omega}_{abc},$$

Using (10), (4.1) and (4.1), we can now set out a clear strategy: first the Ω 's, and then the ω 's. Now let's begin the computation of the Ω 's. Starting with $\hat{\Omega}_{pmn}$:

$$\begin{aligned} \hat{\Omega}_{pmn} &= \frac{1}{2} (\hat{V}_p{}^\mu \hat{V}_m{}^\nu - \hat{V}_m{}^\mu \hat{V}_p{}^\nu) \partial_\nu \hat{V}_{\hat{\mu}n} \\ &= \frac{1}{2} (\hat{V}_p{}^\mu \hat{V}_m{}^\nu - \hat{V}_m{}^\mu \hat{V}_p{}^\nu) \partial_\nu \hat{V}_{\mu n} \\ &= \frac{1}{2} (\hat{V}_p{}^\mu \hat{V}_m{}^\nu - \hat{V}_m{}^\mu \hat{V}_p{}^\nu) \partial_\nu \hat{V}_{\mu n} \\ &= \frac{1}{2} (V_p{}^\mu V_m{}^\nu - V_m{}^\mu V_p{}^\nu) \partial_\nu (V_{\mu n}) \\ &= \frac{1}{2} (V_p{}^\mu V_m{}^\nu - V_m{}^\mu V_p{}^\nu) \partial_\nu (V_{\mu n}) \\ &= \frac{1}{2} (V_p{}^\mu V_m{}^\nu - V_m{}^\mu V_p{}^\nu) \partial_\nu (V_{\mu n}) \\ &= \frac{1}{2} (V_p{}^\mu V_m{}^\nu - V_m{}^\mu V_p{}^\nu) \partial_\nu (V_{\mu n}) \\ &= \frac{1}{2} (V_p{}^\mu V_m{}^\nu - V_m{}^\mu V_p{}^\nu) \partial_\nu (V_{\mu n}) \end{aligned}$$

For $\hat{\Omega}_{pma}$, we find:

$$\begin{aligned} \hat{\Omega}_{pma} &= \frac{1}{2} (\hat{V}_p{}^\mu \hat{V}_m{}^\nu - \hat{V}_m{}^\mu \hat{V}_p{}^\nu) \partial_\nu \hat{V}_{\hat{\mu}a} \\ &= -(V_p{}^\mu V_m{}^\nu - V_m{}^\mu V_p{}^\nu) \partial_\nu (A_\mu^\alpha \Phi_{a\alpha}) + (A_p^\alpha V_m{}^\nu - \hat{A}_m^\alpha V_p{}^\nu) \partial_\nu (\Phi_{a\alpha}) \\ &= -(V_p{}^\mu V_m{}^\nu - V_m{}^\mu V_p{}^\nu) (\partial_\nu (A_\mu^\alpha) \Phi_{a\alpha} + \partial_\nu (\Phi_{a\alpha}) A_\mu^\alpha) + \kappa 1 (A_p^\alpha V_m{}^\nu - \hat{A}_m^\alpha V_p{}^\nu) \partial_\nu (\Phi_{a\alpha}) \\ &= -(V_p{}^\mu V_m{}^\nu - V_m{}^\mu V_p{}^\nu) (\partial_\nu (A_\mu^\alpha) \Phi_{a\alpha} + \partial_\nu (\Phi_{a\alpha}) A_\mu^\alpha) + (A_p^\alpha V_m{}^\nu - \hat{A}_m^\alpha V_p{}^\nu) \partial_\nu (\Phi_{a\alpha}) \\ &= -(V_p{}^\mu V_m{}^\nu - V_m{}^\mu V_p{}^\nu) \partial_\nu (A_\mu^\alpha) \Phi_{a\alpha} \\ &= (F^\alpha_{pm} \Phi_{a\alpha} - \partial_\nu (V_p{}^\mu V_m{}^\nu) A_\mu^\alpha \Phi_{a\alpha} + \partial_\nu (V_p{}^\nu V_m{}^\mu) A_\mu^\alpha \Phi_{a\alpha}) \\ &= F^\alpha_{pm} \Phi_{a\alpha} \end{aligned}$$

For $\hat{\Omega}_{pam}$:

$$\begin{aligned} \hat{\Omega}_{pam} &= \frac{1}{2} (\hat{V}_p{}^\mu \hat{V}_a{}^\nu - \hat{V}_a{}^\mu \hat{V}_p{}^\nu) \partial_\nu \hat{V}_{\hat{\mu}m} \\ &= \frac{1}{2} (\hat{V}_p{}^\mu \hat{V}_a{}^\nu - \hat{V}_a{}^\mu \hat{V}_p{}^\nu) \partial_\nu \hat{V}_{\hat{\mu}m} \\ &= 0 \end{aligned}$$

The same result holds for $\hat{\Omega}_{apm}$.

For $\hat{\Omega}_{pab}$:

$$\begin{aligned}
\hat{\Omega}_{pab} &= \frac{1}{2}(\hat{V}_p^\mu \hat{V}_a^\nu - \hat{V}_a^\mu \hat{V}_p^\nu) \partial_\nu \hat{V}_{\hat{\mu}b} \\
&= \frac{-1}{2} \hat{V}_a^\mu V_p^\nu \partial_\nu \hat{V}_{\hat{\mu}b} \\
&= \frac{1}{2} \Phi_a^\alpha V_p^\nu \partial_\nu \Phi_{\alpha b} \\
&= \frac{1}{2} \Phi_a^\alpha V_p^\nu \partial_\nu (h_{\alpha\beta} \Phi_b^\beta) \\
&= \frac{1}{2} \Phi_a^\alpha \Phi_b^\beta V_p^\nu \partial_\nu h_{\alpha\beta} + \frac{1}{2} \Phi_a^\alpha V_p^\nu h_{\alpha\beta} \partial_\nu \Phi_b^\beta \\
&= \frac{1}{2} \Phi_a^\alpha \Phi_b^\beta V_p^\nu \partial_\nu h_{\alpha\beta} + \frac{1}{2} \Phi_{\beta a} V_p^\nu \partial_\nu \Phi_b^\beta \\
&= \frac{1}{2} \Phi_a^\alpha \Phi_b^\beta V_p^\nu \partial_\nu h_{\alpha\beta} + \frac{1}{2} V_p^\nu \partial_\nu (\underbrace{\Phi_{\beta a} \Phi_b^\beta}_{=\delta_{ab} \text{ i.e is constant } constant}) - \Phi_b^\beta V_p^\nu \partial_\nu \Phi_{\beta a} \\
&= \frac{1}{2} \Phi_a^\alpha \Phi_b^\beta V_p^\nu \partial_\nu h_{\alpha\beta} - \Phi_b^\beta V_p^\nu \partial_\nu \Phi_{\beta a}
\end{aligned}$$

For $\hat{\Omega}_{apb}$:

$$\begin{aligned}
\hat{\Omega}_{apb} &= \frac{1}{2}(\hat{V}_a^\mu \hat{V}_p^\nu - \hat{V}_p^\mu \hat{V}_a^\nu) \partial_\nu \hat{V}_{\hat{\mu}b} \\
&= \frac{1}{2} \hat{V}_a^\mu \hat{V}_p^\nu \partial_\nu \hat{V}_{\hat{\mu}b} \\
&= \frac{1}{2} \hat{V}_a^\alpha \hat{V}_p^\nu \partial_\nu \hat{V}_{\alpha b} \\
&= \frac{-1}{2} \Phi_a^\alpha V_p^\nu \partial_\nu \Phi_{\alpha b}
\end{aligned}$$

For $\hat{\Omega}_{abp}$:

$$\begin{aligned}
\hat{\Omega}_{abp} &= \frac{1}{2}(\hat{V}_a^\mu \hat{V}_b^\nu - \hat{V}_b^\mu \hat{V}_a^\nu) \partial_\nu \hat{V}_{\hat{\mu}p} \\
&= 0
\end{aligned}$$

The same holds for Ω_{abc} . Using (??), we can then obtain the ω 's straightforwardly:

$$\hat{\omega}_{pmn} = \omega_{pmn}, \quad (18a)$$

$$\hat{\omega}_{pma} = F_{pm}^\alpha \Phi_{\alpha a}, \quad (18b)$$

$$\hat{\omega}_{pab} = \frac{1}{2} \Phi_a^\alpha V_p^\mu \partial_\mu \Phi_{\alpha b} - (a \leftrightarrow b), \quad (18c)$$

$$\hat{\omega}_{cmn} = -F_{mn}^\alpha \Phi_{\alpha c}, \quad (18d)$$

$$\hat{\omega}_{cab} = 0 \quad (18e)$$

We can now conclude this part by just computing the multiplication and look at what we got. This give us.

$$\boxed{
\begin{aligned}
S &= \int dx^D \int V \delta \left(\frac{-1}{4} g^{\rho\lambda} \partial_\lambda h_{\alpha\beta} \partial_\rho h^{\alpha\beta} + \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^\beta h_{\alpha\beta} + \right. \\
&\quad \left. + R + \frac{1}{4} \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \log(h) + \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \hat{\phi} \right)
\end{aligned}
} \quad (19)$$

We can observe some interesting things. Firtstly we can see that we have refound the classical Einstein action, coupled with E electromagnetism field. So what we can conclude is that gravity in D+E dimation is gravity in 4d plus E electromagnetism field. and we find a few other field which will come in handy for the dilaton field reduction.

4.5 Adding the dilaton field reduction

For this part we will have to sum $S_{\hat{g}}$ and $S_{\hat{\phi}}$ that gave us

$$\begin{aligned}
S_{\hat{g}+\hat{\phi}} &= \int dx^D \int V \delta \left(\frac{-1}{4} g^{\rho\lambda} \partial_\lambda h_{\alpha\beta} \partial_\rho h^{\alpha\beta} + \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^\beta h_{\alpha\beta} + \right. \\
&\quad \left. + R + \frac{1}{4} \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \log(h) + \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \hat{\phi} + \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\mu}} \hat{\phi} \right)
\end{aligned}$$

We then introduce the shifted dilaton as

$$\phi = \hat{\phi} - \frac{1}{2} \log h = \hat{\phi} - \log \delta \quad (20)$$

wich gave us

$$S = \int dx^D V e^{-\phi} \left(R - \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^{\beta} h_{\alpha\beta} + \right. \\ \left. - g^{\rho\lambda} \partial_{\lambda} h_{\alpha\beta} \partial_{\rho} h^{\alpha\beta} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \phi \partial_{\hat{\nu}} \phi \right) \quad (21)$$

5 Two form fiel reduction

5.1 The reduction

For the two form we will reduce it apart from the two other field, this those not cause any problem as the action is in effect linear.

We can rewrite (5) as

$$\begin{aligned} \mathcal{L}_{\hat{B}} &= \frac{-1}{12} (g^{\hat{\mu}\hat{\mu}'} g^{\hat{\nu}\hat{\nu}'} g^{\hat{\rho}\hat{\rho}'} \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \hat{H}_{\hat{\mu}'\hat{\nu}'\hat{\rho}'}) \\ &= \frac{-1}{12} (\hat{\eta}^{\hat{r}\hat{r}'} \hat{\eta}^{\hat{s}\hat{s}'} \hat{\eta}^{\hat{t}\hat{t}'} \hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{H}_{\hat{r}'\hat{s}'\hat{t}'}) \\ &= \frac{-1}{12} (\hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{H}^{\hat{r}\hat{s}\hat{t}}) \\ &= \frac{-1}{12} (\hat{H}_{rst} \hat{H}^{rst} + \hat{H}_{rsc} \hat{H}^{rsc} + \hat{H}_{rbt} \hat{H}^{rbt} + \hat{H}_{rbc} \hat{H}^{rbc} + \hat{H}_{ast} \hat{H}^{ast} + \hat{H}_{asc} \hat{H}^{asc} + \hat{H}_{abs} \hat{H}^{abs} + \hat{H}_{abc} \hat{H}^{abc}) \end{aligned}$$

however thanks to to the fact that $\hat{B}_{\hat{\mu}\hat{\nu}}$ is antysymmetric one can easily show that this imply that $\hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}}$ is antysemtic for any neighbor index. so we obtaint

$$\mathcal{L}_{\hat{B}} = \frac{-1}{12} (\hat{H}_{rst} \hat{H}^{rst} + 3\hat{H}_{rsc} \hat{H}^{rsc} + 3\hat{H}_{asc} \hat{H}^{asc} + \hat{H}_{abc} \hat{H}^{abc}) \quad (22)$$

and finnaly the action of B is written as

$$S_{\hat{b}} = - \int dx^D \sqrt{-g} e^{-\phi} \left\{ \frac{1}{12} H_{\alpha\beta\gamma} H^{\alpha\beta\gamma st} + \frac{1}{4} H_{\mu\alpha\beta} H^{\mu\alpha\beta} + \frac{1}{4} H_{\mu\nu\alpha} H^{\mu\nu\alpha} + \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right\} \quad (23)$$

Te absence if hat is due to the fact that we define low dimensional form as being:

$$T^{\mu} = V_r^{\mu} \hat{V}_{\hat{\mu}}^r \hat{T}^{\hat{\mu}} \quad (24a)$$

$$T^{\alpha} = V_a^{\alpha} \hat{V}_{\hat{\mu}}^a \hat{T}^{\hat{\mu}} \quad (24b)$$

$$T_{\mu} = V_{\mu}^r \hat{V}_{\hat{\mu}}^r \hat{T}_{\hat{\mu}} \quad (24c)$$

$$T_{\alpha} = V_{\alpha}^a \hat{V}_{\hat{\mu}}^a \hat{T}_{\hat{\mu}} \quad (24d)$$

$$(24e)$$

so with that we obtaint(23)

Now we will compute each componant of non hated H in function of hated H

to begin slowly as we have seen above $\hat{B}_{\alpha\beta} = B_{\alpha\beta}$ and if we use the y independence we get $H_{\alpha\beta\gamma} = 0$

We continue with $H_{\mu\alpha\beta}$

$$\begin{aligned} H_{\mu\alpha\beta} &= V_{\mu}^r \hat{V}_{\hat{\mu}}^r \hat{H}_{\hat{\mu}\alpha\beta} \\ &= \partial_{\mu} B_{\alpha\beta} \end{aligned} \quad (25)$$

those two were quite easy however the next two are going to be more troublesome, but we are going to introduce new object
Now for $H_{\mu\nu\beta}$

$$\begin{aligned} H_{\mu\alpha\beta} &= V_{\mu}^r V_{\nu}^s \hat{V}_{\hat{\mu}}^r \hat{V}_{\hat{s}}^s \hat{H}_{\hat{\mu}\hat{\nu}\hat{\alpha}} \\ &= \hat{H}_{\mu\nu\alpha} - (A_{\mu}^{(1)\alpha} \hat{H}_{\alpha\nu\rho} + A_{\rho}^{(1)\alpha} \hat{H}_{\mu\alpha\nu}) \\ &= \partial_{\mu} \hat{B}_{\nu\alpha} + \partial_{\nu} \hat{B}_{\alpha\mu} - A_{\mu}^{(1)\beta} \partial_{\nu} \hat{B}_{\alpha\beta} - A_{\nu}^{(1)\beta} \partial_{\mu} \hat{B}_{\beta\alpha} \\ &= \partial_{\mu} \hat{B}_{\nu\alpha} + \partial_{\nu} \hat{B}_{\alpha\mu} - \partial_{\nu} (A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta}) + \partial_{\nu} A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta} - \partial_{\mu} (A_{\nu}^{(1)\beta} \hat{B}_{\beta\alpha}) + \partial_{\mu} A_{\nu}^{(1)\beta} \hat{B}_{\beta\alpha} \\ &= \partial_{\mu} \hat{B}_{\nu\alpha} - \partial_{\nu} \hat{B}_{\mu\alpha} - \partial_{\nu} (A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta}) + \partial_{\nu} A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta} + \partial_{\mu} (A_{\nu}^{(1)\beta} \hat{B}_{\alpha\beta}) - \partial_{\mu} A_{\nu}^{(1)\beta} \hat{B}_{\alpha\beta} \\ &= \partial_{\mu} (\hat{B}_{\nu\alpha} + A_{\nu}^{(1)\beta} \hat{B}_{\alpha\beta}) - \partial_{\nu} (\hat{B}_{\mu\alpha} + A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta}) - \hat{B}_{\alpha\beta} (\partial_{\mu} A_{\nu}^{(1)\beta} - \partial_{\nu} A_{\mu}^{(1)\beta}) \\ &= F_{\mu\nu\alpha}^{(2)} - \hat{B}_{\alpha\beta} F_{\mu\nu}^{(1)\beta} \end{aligned} \quad (26)$$

and finally for $H_{\mu\nu\rho}$ we are just going to use the same technique as before which gave us

$$\begin{aligned} H_{\mu\nu\rho} &= V_\mu^r V_\nu^s V_\rho^t \hat{V}_r^{\hat{\mu}} \hat{V}_s^{\hat{\nu}} \hat{V}_t^{\hat{\rho}} \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \\ &= \hat{H}_{\mu\nu\rho} - A_\mu^{(1)\alpha} \hat{H}_{\alpha\nu\rho} - A_\nu^{(1)\alpha} \hat{H}_{\alpha\rho\mu} - A_\rho^{(1)\alpha} \hat{H}_{\alpha\mu\nu} + A_\mu^{(1)\alpha} A_\nu^{(1)\beta} \hat{H}_{\alpha\beta\rho} + A_\nu^{(1)\alpha} A_\rho^{(1)\beta} \hat{H}_{\alpha\beta\mu} + A_\rho^{(1)\alpha} A_\mu^{(1)\beta} \hat{H}_{\alpha\beta\nu} \\ &= \partial_\mu B_{\nu\rho} - \frac{1}{2}(A_\mu^{(1)\alpha} F_{\nu\rho\alpha}^{(2)} + A_{\mu\alpha}^{(2)} F_{\nu\rho}^{(1)}) + \partial_\nu B_{\rho\mu} - \frac{1}{2}(A_\nu^{(1)\alpha} F_{\rho\mu\alpha}^{(2)} + A_{\nu\alpha}^{(2)} F_{\rho\mu}^{(1)}) + \partial_\rho B_{\mu\nu} - \frac{1}{2}(A_\rho^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + A_{\rho\alpha}^{(2)} F_{\mu\nu}^{(1)}) \end{aligned} \quad (27)$$

where

$$B_{\mu\nu} = \hat{B}_{\mu\nu} + \frac{1}{2}A_\mu^{(1)\alpha} A_{\nu\alpha}^{(2)} - \frac{1}{2}A_\nu^{(1)\alpha} A_{\mu\alpha}^{(2)} - A_\mu^{(1)\alpha} B_{\alpha\beta} A_\nu^{(2)\beta} \quad (28)$$

5.2 Observation of gauge

Now as in part 4.2 we are going to observe the gauge rule to be sure that every object transforms as it should so let's begin with $A^{(2)}$ which should transform with the Lie derivative and under the new transformations of $B_{\mu\nu} \rightarrow B_{\mu\nu} + \partial_\mu u \Lambda_\nu$, so let's begin with $A_{\mu\alpha}^{(2)}$

$$\begin{aligned} \delta A_{\mu\alpha}^{(2)} &= \delta(\hat{B}_{\mu\alpha} + B_{\alpha\beta} A_\mu^{(1)\beta}) \\ &= \delta(\hat{B}_{\mu\alpha}) + \delta(B_{\alpha\beta} A_\mu^{(1)\beta}) \\ &= \delta(\hat{B}_{\mu\alpha}) + \delta(B_{\alpha\beta}) A_\mu^{(1)\beta} + B_{\alpha\beta} \delta(A_\mu^{(1)\beta}) \\ &= L_\xi \hat{B}_{\mu\alpha} + \partial_\mu \Lambda_\alpha^{(2)} + A_\mu^{(\beta)} L_\xi \hat{B}_{\alpha\beta} + \hat{B}_{\alpha\beta} \partial_\mu \Lambda_\alpha^{(1)} \\ &= L_\xi \hat{B}_{\mu\alpha} + \partial_\mu \Lambda_\alpha^{(2)} + L_\xi(A_\mu^{(\beta)} \hat{B}_{\alpha\beta}) \\ &= \partial_\mu \Lambda_\alpha^{(2)} + L_\xi A_{\mu\alpha}^{(2)} \end{aligned} \quad (29)$$

so we have the right transformation for this object

Doing the same thing with $B_{\mu\nu}$ we come under the transformation

$$\delta B_{\mu\nu} = \frac{1}{2}(\Lambda^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + \Lambda_\alpha^{(2)} F_{\mu\nu}^{(1)\alpha}), \quad (30)$$

which is exactly the good term to compensate the additional term for the transformation of $H_{\mu\nu\rho}$

6 Complete reduced action

To recap we have the complete reduced action in the form

$$S = \int dx^D V e^{-\phi} \mathcal{L}, \quad (31)$$

Where we decompose S as $S = S_1 + S_2 + S_3 + S_4$, and we pose

$$\begin{aligned} S_1 &= \int dx^D V e^{-\phi} (R + g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi), \\ S_2 &= \int dx^D V e^{-\phi} \left(\frac{1}{4} g^{\mu\nu} (\partial_\mu h_{\alpha\beta} \partial_\nu G^{\alpha\beta} - h^{\alpha\beta} h^{\gamma\delta} \partial_\mu B_{\alpha\gamma} \partial_\nu B_{\beta\delta}) \right), \\ S_3 &= \int dx^D V e^{-\phi} \left(-\frac{1}{4} g^{\mu\rho} g^{\nu\lambda} (h_{\alpha\beta} F_{\mu\nu}^{(1)\alpha} F_{\rho\lambda}^{(1)\beta} + h^{\alpha\beta} H_{\mu\nu\alpha} H_{\rho\lambda\beta}) \right), \\ S_4 &= \int dx^D V e^{-\phi} \left(-\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right). \end{aligned} \quad (32)$$

7 Observation of new symmetry

7.1 O(E,E) group presentation

Now for this part we are going to explore a very important part of physics research nowadays which is symmetry. In our context it will allow us to make equivalence between multiple theories as a whole. But firstly we have to see the obvious symmetry that we have seen in the non-reduced version of the Action which is the $SL(E, R)$. But what will interest us in the reduction is that it does not create nor destroy any symmetry and if we observe a new symmetry in the reduced form that means that we have effectively discovered a new symmetry in the unreduced form. and following the [5] we observe an $O(E, E)$ symmetry. let's define it and then observe if it is true so the definition.

Definition 1 Any matrices M is in $O(E, E)$ if it is a square $E \times E$ matrices and if for the matrices

$$\zeta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

We can use this matrices as a metric for the groupe $O(E, E)$ as it has d -1 eigenvalues and d 1 eigenvalues, in a basis rotated from the one with diagonal metric.

the matrices M verify

$$\boxed{M^T \zeta M = \zeta}$$

Now we will consider that the action of the group on anything that does not have internal index is null so for the metric and ϕ we have

$$g_{\mu\nu} \rightarrow g_{\mu\nu}, \quad (33)$$

$$\phi \rightarrow \phi \quad (34)$$

It is worth noting that for ϕ the whole thing is invariant however each component ($\hat{\phi}$, and $\frac{1}{2}\log(h)$) on there one are not invariant to $O(E, E)$ group, which is why we needed to creatz the shifted dilaton.

We also need to define the action of the groupe on the two form, and for $A^{(2)}$, when they have internal index, so we have:

7.2 Observation of the group

We will now define a new matrices M which is

$$M = \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \quad (35)$$

Of course we will verify that this true

$$\begin{aligned} M^T \zeta M &= \begin{pmatrix} h^{-1} & Bh^{-1} \\ -h^{-1}B & h - Bh^{-1}B \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} Bh^{-1} & h^{-1} \\ h - Bh^{-1}B & -h^{-1}B \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \zeta \end{aligned}$$

so we confirm that $M \in O(E, E)$ Now that we have that in mind let's retok (32) and verify that each $\mathcal{L}$ is infact $O(E, E)$ invariant

For S_1 the invariance is obvious as $g_{\mu\nu}$ and Φ are both invariant. Nonwithsandaning we ought to remamber that for Φ each terms in it are note on there self invariant only.

Now the [5] propose a new tactics to observe the invariance of $\mathcal{L}_2$ which is to rewrite everything in matricial terms. we the obtain

$$S_2 = \frac{1}{4} \text{tr}(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B)$$

And now we are going to do a few manipulation

$$\begin{aligned} S_2 &= \frac{1}{4} \text{tr}(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B) \\ &= \frac{1}{8} \text{tr}(2\partial_\mu h \partial^\mu h^{-1} + 2h^{-1} \partial_\mu B h^{-1} \partial^\mu B) \\ &= \frac{1}{8} \text{tr}(2\partial_\mu h \partial^\mu h^{-1} - 2\partial_\mu (B h^{-1} B) \partial^\mu h^{-1} + \partial_\mu (B h^{-1}) \partial^\mu (B h^{-1}) + \partial_\mu (h^{-1} B) \partial^\mu (h^{-1} B)) \\ &= \frac{1}{8} \text{tr}(\partial_\mu (h - B h^{-1} B) \partial^\mu h^{-1} + \partial_\mu (B h^{-1}) \partial^\mu (B h^{-1}) + \partial_\mu (h^{-1} B) \partial^\mu (h^{-1} B) + \partial^\mu h^{-1} \partial_\mu (h - B h^{-1} B)) \\ &= \frac{1}{8} \text{tr}(\partial_\mu M^{-1} \partial^\mu M), \end{aligned} \quad (36)$$

The $O(E, E)$ then become manifest as M transformation under a global $O(E, E)$ transformation is

$$M \rightarrow \Omega M \Omega^T,$$

where $\Omega \in O(E, E)$.

Now we are going to rewrite $\mathcal{L}_3$ by developing the H partial to get

$$\begin{aligned} S_3 &= -\frac{1}{4} [F_{\mu\nu}^{(1)\alpha} h_{\alpha\beta} F^{(1)\mu\nu\beta} + (F_{\mu\nu\alpha}^{(2)} - B_{\alpha\gamma} F_{\mu\nu}^{(1)\gamma}) h^{\alpha\beta} (F_{\beta}^{(2)\mu\nu} - B_{\beta\delta} F^{(1)\mu\nu\delta})] \\ &= -\frac{1}{4} \mathcal{F}_{\mu\nu}^i (M^{-1})_{ij} \mathcal{F}^{\mu\nu j} \end{aligned} \quad (37)$$

Where $\mathcal{F}_{\mu\nu}^i$ is the 2E-component vector of field strengths

$$\mathcal{F}_{\mu\nu}^i = \begin{pmatrix} F_{\mu\nu}^{(1)\alpha} \\ F_{\mu\nu\alpha}^{(2)} \end{pmatrix} = \partial_\mu \mathcal{A}_\nu^i - \partial_\nu \mathcal{A}_\mu^i \quad (38)$$

Now the O(E,E) invariance is visible if we consider that the vector fields transform linealy to the vector representation of O(E,E), i.e., $\mathcal{A}_\mu^i \rightarrow \Omega_j^i \mathcal{A}_\mu^j$. And finally we go to S_4 firstly let's rewrite $H_{\mu\nu\rho}$

$$\begin{aligned} H_{\mu\nu\rho} &= \partial_\mu B_{\nu\rho} - \frac{1}{2}(A_\mu^{(1)\alpha} F_{\nu\rho\alpha}^{(2)} + A_{\mu\alpha}^{(2)} F_{\nu\rho}^{(1)}) + \partial_\nu B_{\rho\mu} - \frac{1}{2}(A_\nu^{(1)\alpha} F_{\rho\mu\alpha}^{(2)} + A_{\nu\alpha}^{(2)} F_{\rho\mu}^{(1)}) + \partial_\rho B_{\mu\nu} - \frac{1}{2}(A_\rho^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + A_{\rho\alpha}^{(2)} F_{\mu\nu}^{(1)}) \\ &= \partial_\mu B_{\nu\rho} - \frac{1}{2} \mathcal{A}_\mu^i \zeta_{ij} \mathcal{F}_{\nu\rho}^j + (\text{cyc.perm}) \end{aligned} \quad (39)$$

Now as $\Omega^T \zeta \Omega = \zeta$ the second term is O(E,E) invariant. And the first term is invariant if we require that $B_{\nu\rho}$ does not transform. This demonstrates that the action hides a non-trivial symmetry, providing a useful framework for comparing equivalent theories.

8 Conclusion

References

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A lagrangian

By using the principle of least action and the variational principle we can see that

$$\begin{aligned} \delta S &= \delta \int \mathcal{L} dx^\mu \\ &= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho + \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \partial_\nu \eta_\rho dx^\mu \\ &= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho + \underbrace{\partial_\nu \left(\frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho \right)}_{\text{border terme equal to zero}} dx^\mu \\ &= 0 \end{aligned}$$

which gave the equation

$$\boxed{\frac{\partial \mathcal{L}}{\partial \eta_\rho} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} = 0} \quad (40)$$

which is in fact the equation of mouvement for the field

B General relativity 101

Definition 2 Let V be an R -vector space of dimension N , we define the tensor of order $(r, s) \in \llbracket 0, 1 \rrbracket^2$ on $V^{(r,s)} \stackrel{\text{def}}{=} V^{\otimes r} \otimes V^{\otimes s}$ as a being a multilinear application on $V^{*r} \times V^s$ to $\mathbb{R}$

Definition 3 We will call n -dimension ($n \in \mathbb{N}^*$) manifold every Hausdorff topological space $(\mathcal{M}, \tau)$ locally homeomorphic to $\mathbb{R}^n$

Definition 4 Let $(\mathcal{M}, g)$ be a smooth manifold. A metric tensor is a symmetric, non-degenerate bilinear form $g_{\mu\nu}$ on the tangent space $T_p \mathcal{M}$ at each point $p \in \mathcal{M}$, such that

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

In General Relativity we use the signature $(-, +, +, +)$.

Definition 5 In general in RG we use the christofells connexion as it is the sole one wich is mtreci and torsion free.
We define it as being

$$\text{nabla}_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\sigma}^\nu V^\sigma \quad (41)$$

where the christofells symbols are

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \quad (42)$$